Phase 2

Network Defense & Security Infrastructure

Course Name	Network Security
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1. Introduction & Project Scope

Our focus on this phase is to implement the topology and security concepts we intended to do on the previous phase.

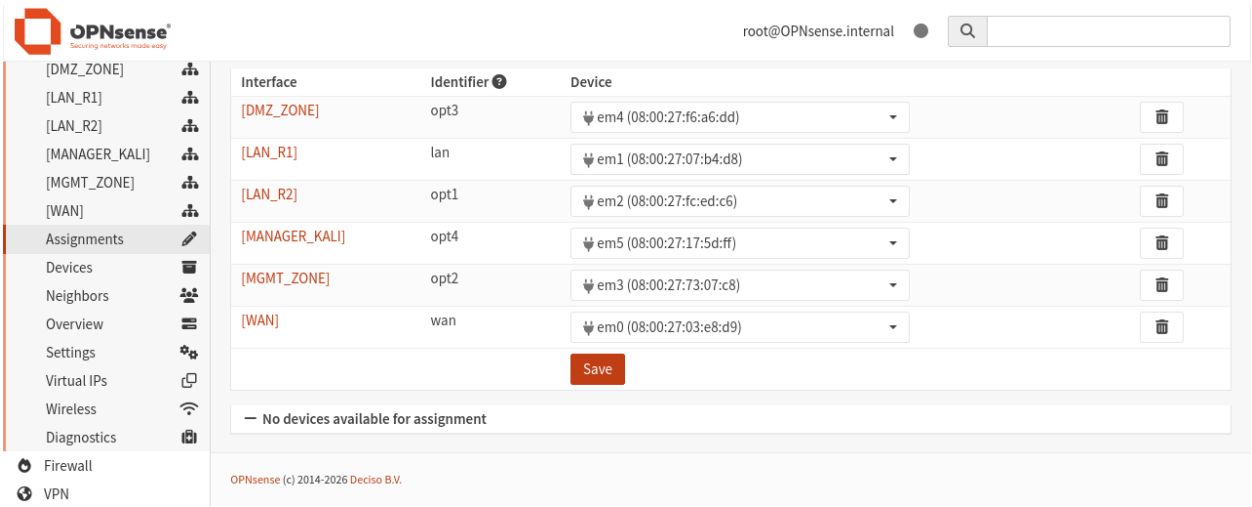
This journey led us from defining our organizational VLANs, into the establishment of a robust edge security layer using OPNsense, through this, we studied about the layering of a network.

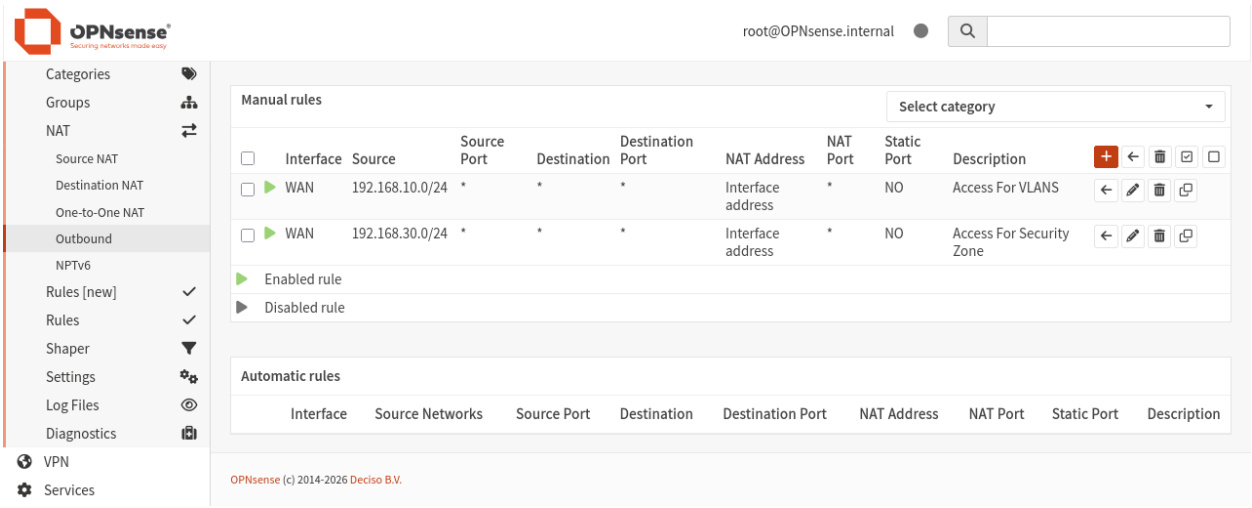
From there, we expanded into the Security Zone (VLAN 30), where we successfully integrated specialized monitoring tools like the Cowrie Honeypot and Splunk SIEM. Our objective was to ensure that every security principle we intended to apply in the previous phase was accurately reflected in thi

2. Network Topology Design.

2.1 Core Security: OPNsense Firewall & VPN

The OPNsense firewall serves as the central hub for the network's security architecture, acting as the primary gateway and dedicated policy enforcement point. It is responsible for managing **Network Address Translation (NAT)** for all external-bound communication, ensuring secure internet access for internal assets.





2.2 Internal Segmentation

To ensure high availability and efficient traffic management, the internal network is segmented across two Layer 3 Switches using a Load Sharing approach, and instead of a traditional Router-on-a-Stick (ROAS) which can create bottlenecks, this project utilizes layer 3 switching with sub-interfaces. This design eliminates a single point of failure and optimizes internal routing

The organizational departments are divided into two logical groups:

Group A: Administrative Departments (L3-Switch-One)

- VLAN 10 (IT), DGW: **`192.168.20.1`**
- VLAN 20 (HR), DGW: **`192.168.21.1`**

Group B: Operational & Financial Departments (L3-Switch-Two)

- VLAN 30 (Finance), DGW: **`192.168.22.2`**
- VLAN 40 (Sales), DGW: **`192.168.23.2`**

2.3 DMZ zone

That is physically and logically isolated from the internal corporate network, it contains a snort, MAIL server, WEB server and cowrie honeypot for the purpose we already explained on the previous phase.

This zone is hosted on a dedicated physical switch connected to a specific interface on the OPNsense firewall to ensure a complete air-gap between public services and private data

2.4 Management zone

This zone, which contains the AAA server and SIEM, is strictly controlled environment, physically and logically isolated to seperate management traffic from general user data.

3. Configuration Details

To ensure clarity and avoid unnecessary duplication, the following command snippets are presented as generic, representative examples for each device category. The configuration breakdown follows a logical "bottom-up" approach based on the project's topology discussed on the previous phase.

3.1 Giving VCPCs IP addresses

```
HR-One> ip 192.168.20
HR-One> save
```

3.2 Configure access layer switch

```
! We intended to use these commands to create vlans, but due to !
problems on saving the vlans on vlan.dat we avoided this and
! used the built-in ethernet switch with GUI config,not commands
AcessLayerSwitch(config)# vlan 10
AcessLayerSwitch(config-vlan)# name HR
AcessLayerSwitch(config-vlan)# exit
AcessLayerSwitch(config)# int fa0/1
AcessLayerSwitch(config-if)# switchport mode access
AcessLayerSwitch(config-if)# switchport access vlan 10
AcessLayerSwitch(config-if)# exit
! So at conclusion, that code was avoided
```

3.3 Configure layer 3 switch

```
! First we give ip addresses for all interfaces
L3-Switch(config)# ip routing
L3-Switch(config)# int gi0/2.10
L3-Switch(config-subif)# encapsulation dot1Q 10
L3-Switch(config-subif)# ip address 192.168.20.1 255.255.255.0
L3-Switch(config-subif)# no shutdown
L3-Switch(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.5
```

3.4 Configure Router

```
! Looks much as layer 3 switch configuration
Router(config)# int gi0/0
Router(config-if)# no shutdown
Router(config-if)# ip address 10.0.0.6 ! point-to-point mask
Router(config-if)# exit
Router(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.5
! point-to-point link is prefered to be used between routers connection
```

3.5 Configure OPNSense

```
! That looks a bit different, it is not commands actually it is
question and answer with the system, leads to what we need, the
steps are:
1. Login (using root and password) --> that will open a menu from
   1 to 12
2. Assigning interfaces (press 1 to enter that), our assignment
   was
a. Wan                    -> em0 --->        using DHCP ip
b. Lan_R1                 -> em1 --->        static 10.0.0.1/30
c. DMZ                    -> em2 --->        static 10.0.0.5/30
d. Lan_R2                 -> em3 --->        static
   192.168.100.1/24
e. Management-Zone        -> em4 --->        static
   192.168.30.1/24
f. Manager                -> em5 --->        static
   192.168.50.1/24
3. Set ip addresses (press 2 to enter that)
```

3.6 Configure snort

```
# First define HOMENET, EXTERNAL_NET
alert tcp $EXTERNAL_NET any -> $HOME_NET 22 (msg:"ET SCAN
Potential SSH Brute Force"; flags:S; threshold:type threshold,
track by_src, count 5, seconds 60; sid:1000001; rev:1
# That is used agains ssh, it is just an example of a rule
# On the context of our project, we sure connected these alerts
to SIEM
```

3.7 Configure honeypot

```
# We must first set an ip address, that is by entering the .yaml
file
sudo nano /etc/netplan/00-installer-config.yaml
# We may need the sudo netplan apply after any change on these
files
# Please be aware that this file is not only for the honeypot, we
# already used it with the WEB, MAIL servers and others.

# Install dependencies (critical pkgs needed)
sudo apt update && sudo apt install -y python3-venv libssl-dev
libffi-dev build-essential python3-pip git

# Now we need to put a user for the cowrie, the purpose is not to
give #the hacker so many permissions,not the permissions root,
e.g. , for #example
sudo adduser --disabled-password cowrie

#crete a virtual enviroment, to make sure not to ruin the real
one
python3 -m venv cowrie-env

# install cowrie
pip install --upgrade pip
pip install -r requirements.txt

# start cowrie
bin/cowrie start

#verify cowrie
bin/cowrie status

#cowrie configuration, and that needs explaination, we must
change the
#hostname and the listen_port, for example
hostname = webserver
listen_port = 2222 # insted of ssh

#what happens here is that the hacker reads the hostname as
#webserver which is useful for his purposes, and then when he try
#to ssh this, which is connecting the port 22, it will transfer
#him to 2222 without him knowing, and we already set a user with
#minimal permissions and a virtual environment for him
```

3.8 Configure MAIL & WEB servers

```
#reach to the .yaml file we discussed on the honeypot
configuration
#and do the routine things

#install apache
sudo apt update && sudo apt install -y apache2

#verify apache is running
sudo systemctl status apache2

#enable any service
sudo systemctl enable [service_name]
```

3.9 Configure AAA server

```
#do the routine things we discussed about the .yaml file

#we used freeRadius as the AAA server, to install it:
sudo apt update && sudo apt install -y freeradius

#verify it is running
sudo systemctl status freeradius

#add a user, on the file /etc/freeradius/3.0/users, for example:
Admin   Cleartext-Password := "admin123"
        Reply-Message := "Welcome to the network"

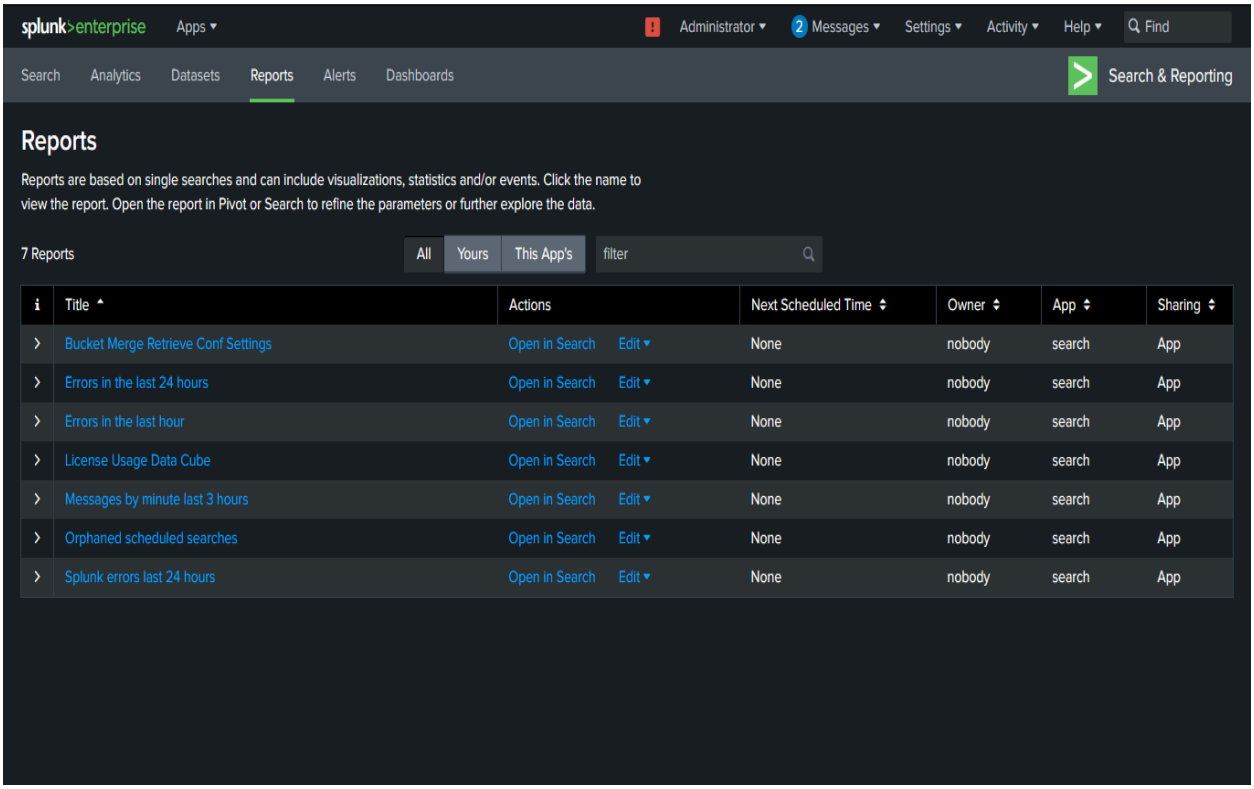
#add the network as a client, for example:
client dmz_zone {
   ipaddr = 192.168.10.0/24
   secret = radius123
   shotname = dmz
}
```

3.10 Configure Splunk server

```
#do the routine things we discussed about the .yaml file

#download splunk which will be used as SIEM
wget -O splunk-9.2.1-78803f08cc4d-linux-2.6-amd64.dev
'https://download.splunk.com/products/splunk/releases/9.2.1/linux
/splunk-9.2.1-78803f08cc4d-linux-2.6-amd64.deb'

#still working on its configuration significant milestones have
already #been achieved, including core installation and
environment staging, #will provide the full configuration on the
final phase, to make sure #not to configure something now and
change it later
```



4. Testing & Final notes

Current Status:

We've already covered a lot of ground here. Most of the core features are up and running, and everything looks solid so far. We are still fine-tuning some of the advanced SIEM alerts.

We made sure and tested using the classic ping way, services checks, etc.

Before wrapping up, here are some important details about the environment we built in GNS3:

- Management via Internal Manager: We didn't use Windows for management. Instead, we used an internal linux manager (Kali GUI) to write all the firewall rules for OPNsense and to configure the Splunk dashboard.

  This kept the environment lightweight and consistent.

- Device Images:

  Layer 3 Switches & Routers: These were high-performance QEMU images to ensure we have full Cisco IOS functionality.

  Access Layer Switches: We used the Built-in GNS3 switches for simple port distribution.

  The VirtualBox & Ubuntu Edge: The majority of our servers (Web, Mail, Splunk, Cowrie) were running as VirtualBox machines.

  The Clone Trick: To save time and resources, we heavily relied on the Linked Clone feature. We built one solid, updated Ubuntu Server base image and then cloned it for each specific role. This ensured all our servers had the same core stability while we only changed the specific configurations (like installing Postfix on one and Snort on another).